

Programme	B. Sc. Statistics				
Course Code	STA5EJ302				
Course Title	Optimization Techniques				
Type of Course	Major Elective				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary Objective	Make students to formulate, solve and implement feasible solutions of complex Industrial, Trade, Commercial problems				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding Basics of Operations Research.	U	C	Instructor-created exams / Quiz
CO2	Distinguishing Solution, Feasible Solution, Basic Solution and Basic Feasible Solutions	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Mathematical Formulation Real life problems	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Mastering Simplex Algorithm	U	C	Instructor-created exams / Home Assignments
CO5	Apply LPP in Transportation and Assignment Problems	Ap	P	One Minute Reflection Writing assignments
CO6	Analyse decision making under conflict	Ap	P	Viva Voce

	Game theory			
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Liner Programming Problem		14	20
	1	Graphical Solution of LPP.		
	2	Feasible Solution, Basic Feasible Solution of LPP		
	3	Simplex Algorithm without Artificial Variables.		
	4	Artificial Variable technique		
	5	Big M method		
	6	Two Phase method		
	Sections from References:			
II	Application of LPP		10	15
	7	Duality Primal and Dual LPP		
	8	Economic Interpretation of Dual		
	9	Dual Simplex Method Solution of primal using Dual.		
	10	Transportation and Assignment Problems as special case of LPP.		
	11	Balanced Transportation Problem, Balanced Assignment Problem		
	12	Initial Basic Feasible Solution using NWCR		
	13	Initial Basic Feasible Solution using LCCM		
	Sections from References:			
III	Solving TP & AP		12	20
	14	Solution of Transportation Problem using Vogel's Approximation Method		
	15	Optimization using MODI Method		

	16	Hungarian Method of Solving Assignment Problem		
	Sections from References:			
IV	Game Theory			
	17	Decision making under Conflict		
	18	Pay off Matrix.		
	19	MinMax MaxMin Criteria		
	20	Pure and Mixed Strategy		
	21	Value of Game and Saddle Point		
	22	Principle of Dominance, solving 2x2 games.		
	23	Graphical solution of 2xn and nx2 games		
	Sections from References:			
V	Open Ended Module:		12	15
	1	Origin, Development of OR. Nature & Scope of OR, Uses & Limitations of OR. Linear Programming Problem, Mathematical Formulation, General, Standard form of LPP.		
	Sections from References:			
Books and References:				
1. Operations Research, Swaroop, Kanti, P. K. Gupta and Man Mohan. 2007. 13th Edition. New Delhi: Sultan Chand and Sons				
2. Operations Research, J K Sharma, Laxmi Publications				
3. Operations Research V K Kapoor Sulthan Chand and Sons				